

GreenTech Solutions – JavaScript Application Task

This assignment is marked out of 20.

Purpose of this assignment

In this assignment, you will **apply the JavaScript skills learned in class** to improve the GreenTech Sustainability Tracker.

You will:

- work independently
- build on the starter code
- improve user experience and functionality
- demonstrate clear, structured JavaScript

This assignment is designed to prepare you for **Task 2 – Developing the Solution**.

What you are given

You must continue using:

- the same project folder
- the same index.html
- the same style.css
- your existing app.js

You must **not start again**.

What you must complete

You will focus on **making the GreenTech section work properly**, using the JavaScript patterns already learned.

Task requirements (what to build)

1. Character counter (core requirement)

- The forum text box must update the character count as the user types
- This must work reliably

This should reuse the same pattern as the practice area.

2. Clear form behaviour (core requirement)

When the **Clear form** button is clicked:

- the form resets
- all messages are cleared
- impact score resets to “No score yet”
- eco tip resets to “No tip yet”
- character count resets to 0

3. Form submission handling (core requirement)

When **Calculate impact** is clicked:

- the page must not refresh
- a message must appear to confirm submission
- JavaScript must control the behaviour

4. Input handling and validation (core requirement)

You must validate **at least two** of the following:

- name is not blank
- travel mode is selected
- car miles is 0 or more
- energy used is 0 or more

If input is invalid:

- show a clear message
- stop the rest of the code from running

5. Results section (application requirement)

You must update **both**:

- Impact score
- Suggested eco tip

These must change based on user input.

The logic can be simple, but it must be clear and sensible.

Suggested logic (you may adapt this)

Example approach:

- If travel mode is car:
 - higher impact score
 - tip about reducing car use
- If travel mode is not car:
 - lower impact score
 - positive reinforcement message

You do **not** need a complex formula.

Code quality expectations

Your JavaScript should:

- use clear variable names
- be split into logical sections
- include comments explaining what the code does

- reuse patterns from the lesson
- avoid repeated or messy code

Testing evidence (required)

You must complete **docs/testing-log.md**.

Include:

- one normal test
- one blank or missing input test
- one invalid or extreme input test

Explain briefly what happened and what you changed.

Marking (20 marks total)

Your work will be marked using the bands below.

Band 1: Basic attempt (1–7 marks)

- JavaScript runs with limited success
- Some interactivity works, but not reliably
- Validation is minimal or inconsistent
- Results section may show static or unclear output
- Code structure is basic

This shows an early understanding of JavaScript.

Band 2: Secure application (8–14 marks)

- JavaScript works reliably
- Form submission is handled correctly
- Validation is clear and prevents errors

- Impact score and eco tip update meaningfully
- Code is mostly clear and commented
- Testing log is completed sensibly

This shows a solid understanding and correct application.

Band 3: Confident and well-structured (15–20 marks)

- All required features work correctly
- Validation is clear and user-friendly
- Results section updates logically and clearly
- JavaScript is well-organised and commented
- Code is easy to follow
- Testing log shows thoughtful testing and fixes
- User experience is clearly considered

This shows strong preparation for Task 2.

Submission checklist (use before handing in)

Before submitting, check:

- All JavaScript runs without errors
- The page does not refresh on submit
- Invalid input shows helpful messages
- Impact score and eco tip update correctly
- testing-log.md is completed
- Code is commented and readable

Final reminder

You are **not expected to be perfect**.

You are expected to:

- apply what you learned
- explain your logic
- show progress and effort

If you can explain your JavaScript, you are doing it right.